TAYTEN BENNETSEN

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EDUCATION

Rice University — Houston, Texas

B.S. in Computer Science and B.A. in Cognitive Science — Expected Graduation, April 2027

- **GPA:** 3.80/4.00
- **Concentrations:** Data Structures and Algorithms, Software Engineering, Machine Learning, OOP, Operating Systems

EXPERIENCE & PROJECTS

Software Engineer Intern | Wostmann & Associates — May 2024 - Aug 2024

Created open source screen reader for visually impaired users — C++ | WinAPI | Microsoft UI Automation | Windows Speech API

- Designed **SightSpeak Reader** using C++, integrating WinAPI, Microsoft UI Automation, and Windows Speech API to enable real-time text-to-speech output and visual text highlighting, supporting over **4,000** Microsoft applications.
- Developed **8** efficient keyboard shortcuts and utilized Breadth-First Search (BFS) tree traversal for seamless navigation through UI elements, improving navigation speed by **80%** for beta users.
- Leveraged a **thread pool** and **futures/promises** to manage multiple tasks concurrently, reducing text processing latency by **28ms** and ensuring a smooth user experience.
- Implemented **Tesseract OCR** to enable text extraction from images and PDFs, achieving a **98%** success rate.

Full-Stack Project | Pillora – Medication Management Mobile Application — Sep 2024 - Oct 2024

Developed during HackRice hackathon — JavaScript | React | Node.js | MongoDB | Tailwind | OpenAI | OpenFDA | Express

- Led a team of **4** to build a cross-platform mobile app using **React** and **Node.js**, targeting an increase in medication adherence.
- Implemented **Retrieval-Augmented Generation (RAG)** by integrating the **OpenFDA API** with **OpenAI**, enhancing the AI-driven chatbot's accuracy by **35%** and reducing hallucinations by **90%** during internal testing.
- Implemented secure **JWT-based authentication**, ensuring data protection for user accounts and real-time notifications across multiple devices.
- Architected a scalable backend using **MongoDB** and **Express**, capable of handling up to **250** concurrent users, laying the groundwork for future feature expansions and a projected user base growth of **200%**.
- Recieved **2nd place overall** and **1st in the healthcare path** at **HackRice 2024**, recognizing excellence in software innovation.

Full-Stack Project | Rice Dance Theatre Ticketing System — Dec 2023 - Apr 2024

Ticketing website and mobile application — JavaScript | React | Node.js | MongoDB | CSS | HTML | Git | npm

- Collaborated with a team of **15** to design and implement an intuitive user interface using **JavaScript** and **React**, projected to handle up to **1,500** monthly users for seamless public ticket purchasing and admin event management.
- Integrated secure authorization using **Auth0** and payment gateways to facilitate online transactions, ensuring a user-friendly and secure experience.
- Led the design and construction of the backend system using **Node.js** and **MongoDB**, ensuring scalable and efficient data management to support high traffic and secure transactions.
- Implemented features such as seat selection, real-time availability updates, and personalized user accounts to enhance the user experience.

AI LangChain Project | Enhanced Security Assistant — Oct 2024 - Nov 2024

Vulnerability analysis and security research tool — Python | LangChain | OpenAI API | NVD API | Pydantic | React

- Developed a comprehensive security research tool integrating the **National Vulnerability Database (NVD) API** with **LangChain** and **OpenAI**, enabling natural language queries for vulnerability analysis across **4** distinct tool functions.
- Implemented robust error handling and rate limiting, achieving **99.9%** uptime during testing and reducing API failures by **95%** through intelligent request management.
- Designed a modular architecture using **Pydantic** for type safety and validation, decreasing input-related errors by **85%** and improving code maintainability.
- Created an intuitive chat interface handling **3+** concurrent operations, with response times averaging under **2** seconds for complex queries spanning multiple data sources

Rice Apps Dev | Software Engineering Club — Sep 2023 - Present

- Participated in weekly Sprints, refining software engineering skills through iterative development and problem-solving.
- Used JavaScript, React, CSS and HTML to create a Portfolio, Shopping List API and multiple client projects.

TECHNICAL SKILLS

Programming | C/C++, React, React Native, JavaScript, Java, Pandas, NodeJS, NextJS, CSS, HTML, Python, MongoDB, SQL, R
Tools | Postman, Expo, Chai, Git, npm, Docker, Visual Studio Code, Jupyter Notebooks, Google Colab, OpenAI, Auth0, OCR, LangChain